Neff's α in the Pauli–Villars multigrid of Möbius domain-wall fermions: measurements

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1 Question

Neff's form of the domain-wall matrix with $\alpha = 0.6$ reduces the iterations of even-odd CG substantially. In HMC on 24^4 at $m = 0.001$ (setup S below), mixed-precision CG takes 31996 iterations at $\alpha = 0.6$ against 40912 at $\alpha = 1$. The multigrid of RutticeQCD gains little: it takes 362 against 356 in the same runs. On $24^3 \times 48$ the gain also depends on the setup (Table 1). We would like to understand where α acts in the multigrid, and whether the coarse space should be built for $D^{(\alpha)}$ instead of D.

2 Operator and multigrid

Neff's form is $D^{(\alpha)} = DQ$ with Q diagonal in the fifth dimension,

$$Q_0 = P_- + \alpha P_+, \qquad Q_s = \alpha \;\; (0 < s < L_s - 1), \qquad Q_{L_s-1} = P_+ + \alpha P_-. \tag{1}$$

Q leaves $P_-\psi_0$ and $P_+\psi_{L_s-1}$ unchanged, which make up the quark field, so the four-dimensional propagator is that of D. Up to the overall factor α, which a Krylov method does not see, $\alpha < 1$ weights the quark components on the walls by $1/\alpha$ relative to the rest. The outer solver is flexible GCR keeping two directions, on the left even-odd Pauli–Villars form with α in D only,

$$\tilde{A}_L^{(\alpha)} = (C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger \, Q_e^{-1} \, (C_e^{-1} S) \, Q_e, \tag{2}$$

with S and S_{PV} the even-odd Schur complements of D and D_{PV} and C_e, $C_{\mathrm{PV},e}$ their site-diagonal parts on the even sites. The preconditioner is a two-grid cycle $M =$ Chebyshev smoother, coarse correction E, Chebyshev smoother. Its parts are built as follows:

part of M	operator it is built from
test vectors, P (filter and bootstrap)	plain D, D_{PV}
coarse operators $P^\dagger D P$, $P^\dagger D_{\mathrm{PV}}^\dagger P$	plain D, D_{PV}
site-local factors $O_R = 1$, $O_M = C_e Q_e C_{\mathrm{PV},e}^\dagger$, $O_L = Q_e^{-1}$ of E	$\tilde{A}_L^{(\alpha)}$
Chebyshev smoother and the top λ_{hi} of its interval	$\tilde{A}_L^{(\alpha)}$

The correction follows from $\tilde{A}_L^{(\alpha)\,-1} = Q_e^{-1} S^{-1} C_e Q_e C_{\mathrm{PV},e}^\dagger (S_{\mathrm{PV}}^\dagger)^{-1}$:

$$E = O_L \, \pi P (P^\dagger D P)^{-1} P^\dagger \iota \; O_M \; \pi P (P^\dagger D_{\mathrm{PV}}^\dagger P)^{-1} P^\dagger \iota \; O_R, \tag{3}$$

with ι and π the embedding of the even sites and the projection onto them. Every α therefore shares one setup. For the right form the correction is exact on the vectors P is built from; for the left form it is not, already at $\alpha = 1$.

Setups. Möbius $(b, c) = (1.5, 0.5)$, $L_s = 8$, $m_{\mathrm{PV}} = 1$, $\alpha_{\mathrm{PV}} = 1$. Setup S: $M_5 = 1.0$, stout links ($\rho = 0.1$, 3 steps), Chebyshev with 8 steps on $[\lambda_{\mathrm{hi}}/10, \lambda_{\mathrm{hi}}]$. Setup B: $M_5 = 1.8$, unsmeared links, 12 steps on $[\lambda_{\mathrm{hi}}/30, \lambda_{\mathrm{hi}}]$. Aggregates of 4^4 sites with the whole fifth dimension, 32 test vectors (16 filtered low modes, each lifted to two columns), coarse solves in single precision to $|r|^2/|b|^2 < 10^{-2}$, GCR(2) to 10^{-18}. Configurations: quenched, plaquette action at $\beta = 6.0$. The 12^4 one has one unpaired near-zero mode ($Q = +1$). All runs on one GPU.

3 Iterations

Table 1: GCR(2) iterations with the multigrid, $\alpha = 0.6$ / $\alpha = 1$.

	12^4		$24^3 \times 48$	
m	S	B	S	B
0.1	8/9	7/9	13/16	7/9
0.01	–	–	15/15	15/17
0.001	14/13	14/15	28/27	26/28

In setup B, $\alpha = 0.6$ saves one to three iterations at every mass on both volumes. In setup S it saves at $m = 0.1$ and loses one at $m = 0.001$. The 12^4 lattice shows the same pattern as $24^3 \times 48$, so the measurements below are on 12^4.

4 Residual histories

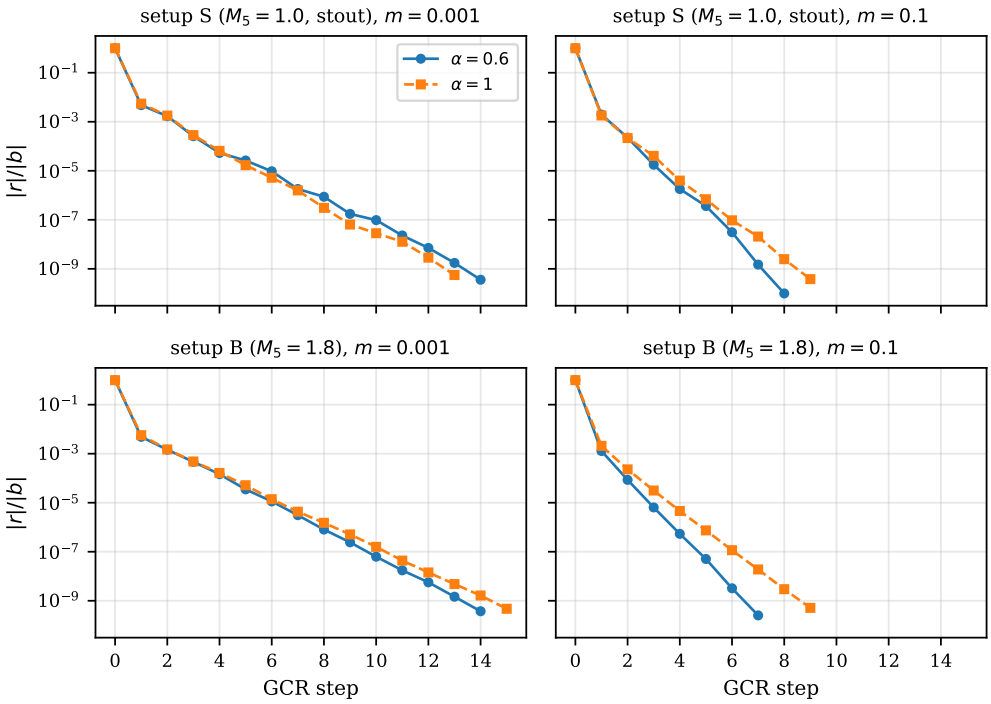


Figure 1: GCR(2) residual histories on 12^4, one random right-hand side.

The difference comes from the rate after the first cycle (Fig. 1, Table 2). At $m = 0.1$, $\alpha = 0.6$ converges faster from the second step on, by a factor of two per step in setup B. At $m = 0.001$ the first

Table 2: Mean reduction of $|r|/|b|$ per GCR step, $\alpha = 0.6$ / $\alpha = 1$, over steps 1–4 and from step 4 to convergence. The first step reduces $|r|/|b|$ to 1–6 $\times 10^{-3}$ in every case.

setup	m	steps 1–4	step 4 to end
S	0.001	0.224/0.228	0.304/0.274
S	0.1	0.097/0.130	0.086/0.157
B	0.001	0.310/0.306	0.276/0.313
B	0.1	0.075/0.130	0.078/0.162

four steps agree within a few percent in both setups. Setup S at $m = 0.001$ then slows down with $\alpha = 0.6$, with several steps that reduce the residual by only 0.36–0.56, once $|r|/|b|$ is below about 10^{-4}.

5 Spectra

$\tilde{A}_L^{(\alpha)}$ **without the multigrid.** Implicitly restarted Arnoldi, 48 eigenvalues of smallest real part and 4 of largest magnitude:

setup	m	$\lvert\lambda_0\rvert$, $\alpha = 0.6/1$	$\lvert\lambda_{47}\rvert$	$\lvert\lambda\rvert_{\max}$
S	0.001	$3.72/2.32 \times 10^{-4}$	0.155/0.096	1.981/1.980
S	0.1	$3.63/2.29 \times 10^{-2}$	0.151/0.099	1.981/1.980
B	0.001	$1.49/1.05 \times 10^{-3}$	0.118/0.082	4.442/4.454
B	0.1	$4.10/3.04 \times 10^{-2}$	0.074/0.082	4.442/4.453

$\alpha = 0.6$ raises the smallest eigenvalue by a factor 1.35–1.6 and leaves the largest unchanged to 0.3%. The factor is 1.6 in S at both masses and 1.42, 1.35 in B.

$\tilde{A}_L^{(\alpha)} M$**.** The same Arnoldi on the preconditioned operator, with M built as in the solver but with exact coarse solves (double precision, 10^{-20}), so that M is linear. κ is the condition number $1/|y^\dagger x|$ of λ_0, from the left and right eigenvectors; $\|I - \tilde{A}_L S\|_2$ and $\|I - \tilde{A}_L E\|_2$ are those of the smoother and of the correction alone.

setup	m	α	$\mathrm{Re}\,\lambda_0$	$\kappa(\lambda_0)$	$\lvert\lambda\rvert_{\max}$	$\|I - \tilde{A}_L S\|_2$	$\|I - \tilde{A}_L E\|_2$
S	0.001	0.6	0.499	17.8	4.27	1.038	2.83×10^4
		1	0.863	8.1	5.48	1.022	1.90×10^4
S	0.1	0.6	0.865	8.5	8.08	0.674	3.29×10^4
		1	0.902	8.0	6.09	0.776	2.22×10^4
B	0.001	0.6	0.687	9.2	1.71	1.173	3.87×10^3
		1	0.830	15.3	1.75	1.117	2.81×10^3
B	0.1	0.6	0.952	11.6	1.30	0.677	1.22×10^3
		1	0.913	6.1	1.28	0.779	9.13×10^2

The spread $|\lambda|_{\max}/\mathrm{Re}\,\lambda_0$ is larger with $\alpha = 0.6$ in three cases and unchanged in B at $m = 0.1$, where $\alpha = 0.6$ gains the most. The eigenvalues of this linear $\tilde{A}_L^{(\alpha)} M$ therefore do not follow the convergence rates of Table 2. At $m = 0.1$, $\alpha = 0.6$ reduces $\|I - \tilde{A}_L S\|_2$ by 13% in both setups. With $\alpha = 0.6$, $\|I - \tilde{A}_L E\|_2$ is larger by a factor 1.34–1.49 in all four cases.

6 The cycle on the low modes of $\tilde{A}_L^{(\alpha)}$

For the 8 eigenvectors x of $\tilde{A}_L^{(\alpha)}$ of smallest real part ($|x| = 1$, eigenpair residuals below 4×10^{-8}) we measure

$$\delta_X(x) = \min_c |x - c\,\tilde{A}_L^{(\alpha)} X x|, \qquad X = E,\ M, \tag{4}$$

the part of $\tilde{A}_L^{(\alpha)} X x$ off the direction of x. GCR chooses the scale c itself, so δ_X is the quantity that matters for one step: $\delta = 0$ when X inverts $\tilde{A}_L^{(\alpha)}$ on x, $\delta = 1$ when $\tilde{A}_L^{(\alpha)} X x \perp x$. With $\alpha = 0.6$ the cycle treats

Table 3: δ_M over the 8 lowest eigenvectors, median (largest), with exact coarse solves and with the solver's 10^{-2}.

		exact		10^{-2}	
setup	m	$\alpha = 0.6$	$\alpha = 1$	$\alpha = 0.6$	$\alpha = 1$
S	0.001	0.917 (0.994)	0.620 (0.977)	0.990 (0.995)	0.958 (0.979)
S	0.1	0.918 (0.977)	0.737 (0.882)	0.972 (0.993)	0.889 (0.966)
B	0.001	0.788 (0.919)	0.668 (0.887)	0.828 (0.919)	0.689 (0.887)
B	0.1	0.339 (0.398)	0.340 (0.407)	0.366 (0.454)	0.364 (0.449)

the low modes worse in three of the four cases, by 0.30, 0.18 and 0.12 in the median with exact solves (S at $m = 0.001$ and 0.1, B at $m = 0.001$), and the same in B at $m = 0.1$. The loss is larger in setup S. The correction alone gives $\delta_E = 0.98$–1.00 in every case, at $\alpha = 1$ too: on these eigenvectors $\tilde{A}_L^{(\alpha)} E x$ is nearly orthogonal to x. The reduction of these modes in one cycle, where there is any, comes from the full cycle.

7 A reading

The data are consistent with two effects of α of opposite sign. This reading is a hypothesis; the two effects have not been separated in a solve.

1. The smoother works on $\tilde{A}_L^{(\alpha)}$, and on the part of the spectrum it covers $\alpha = 0.6$ helps it ($\|I - \tilde{A}_L S\|_2$ lower by 13% at $m = 0.1$).

2. P and the coarse operators are those of D, and with $\alpha = 0.6$ the cycle treats the low modes of $\tilde{A}_L^{(\alpha)}$ worse (Table 3).

At $m = 0.1$ the low modes do not set the cost, and the first effect gives the faster rate in both setups. At $m = 0.001$ the low modes matter. In B the smoother interval is wide and the loss on the low modes is small, so the net effect is still a gain. In S the interval is narrow, the coarse correction carries the low modes, the loss is largest, and α costs an iteration. The late slowdown of S at $m = 0.001$ in Fig. 1, after the residual has fallen below 10^{-4}, fits a remainder made of low modes.

8 Questions

1. Should the coarse space be built from the near-null vectors of $D^{(\alpha)}$, which are Q^{-1} times those of D, when the outer operator carries α? Our aggregates keep the whole fifth dimension and split it only by Γ_5 chirality, so we expect $Q^{-1}y$ to lie outside the range of P in general; we have not checked this. A P built from $Q^{-1}y$ has not been tried.

2. Is $\delta_E \approx 1$ on the low eigenvectors expected for the left form, with only the smoothed cycle reducing them?

3. For comparison, an earlier Bridge++ implementation uses the same $D^{(\alpha)} = DQ$, in the whole-lattice operator $(D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^{\dagger} C^{-1} D$ with C the site-local block, and forms the coarse operator from the operator with α. On a 12^4 cold configuration at $\beta = 5.7$ and $m = 0.005$, $\alpha = 0.6$ raised $\lambda_{\max}$ there from 4.66 to 10.6, and its multigrid took 266 iterations against 210 at $\alpha = 1$. Here $\lambda_{\max}$ of $\tilde{A}_L^{(\alpha)}$ does not change with α. Would one expect the difference from the whole-lattice form against the even-odd left form, from the coarse space, or from the configuration? One candidate for the form: $\tilde{A}_L^{(\alpha)}$ is similar to $S_{\mathrm{PV}}^{\dagger} C_{\mathrm{PV}}^{\dagger\,-1} Q_e^{-1} C_e^{-1} S$, with the site-local factors between the two Schur complements, while in the whole-lattice operator Q^{-1} multiplies $C^{-1} D$ from one side only.

4. With α in D_{PV} as well (the code of 2026-10-11, 12^4, the same setup and coarse factors), the multigrid took 25 GCR(2) cycles at $\alpha = 0.6$ (three right-hand sides) and 58 at $\alpha = 0.5$ (one right-hand side); FGMRES(16) took 45–46 at $\alpha = 0.5$. With α in D only it took 24 at $\alpha = 0.5$. What makes the cycle fail there?

Data. The tables and the figure script are in `docs/alpha_study/`.